

Standard Output and Input

In Java, `System.out` and `System.in` are the primary tools used to communicate with a user through the console or command prompt.

Java has 3 streams called `System.in`, `System.out`, and `System.err` which are commonly used to provide input to, and output from Java applications. They handle data traveling in two opposite directions: **sending information out** to the screen and **bringing information in** from the keyboard.

Study link:

<https://jenkov.com/tutorials/java-io/system-in-out-error.html>

System.out (Standard Output)

`System.out` is used to send data **out** of your program and print it onto the screen.

- **How it works:** `System` is a built-in Java class, and `out` is an object inside it representing your computer screen.
- **The Tools:** It has two main functions you will use constantly.

System.out.println() (Print Line) [1] Prints whatever text you give it, and then moves the cursor to a brand-new line	<code>System.out.println("Hello");</code> <code>System.out.println("World");</code> // Output on screen: // Hello // World
System.out.print() (Print Only) [1] Prints the text but keeps the cursor on the exact same line . Anything printed next will appear right beside it.	<code>System.out.print("Hello ");</code> <code>System.out.print("World");</code> // Output on screen: // Hello World

System.in (Standard Input)

System.in is used to bring data **into** your program from the outside world (usually by reading what the user types on their keyboard).

- **The Catch:** By itself, System.in is very basic. It only reads typing in raw computer bytes, which is hard for humans to work with.
- **The Fix:** To make it useful, we pass System.in into a helper tool called a **Scanner**. The Scanner takes those raw bytes and translates them into words or numbers we can easily use.

How to use it in code:

Common Scanner Tools for System.in

Once you set up your scanner, you can read different types of user data using these commands:

- scanner.nextInt() — Reads a whole whole number (like 25).
- scanner.nextDouble() — Reads a decimal number (like 19.99).
- scanner.next() — Reads a single word (stops at the first space).
- scanner.nextLine() — Reads an entire sentence or line of text.

```
// 1. Tell Java you want to use the Scanner tool (put this at the very top of your file)
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        // 2. Create the Scanner and hook it up to System.in
        Scanner keyboard = new Scanner(System.in);

        System.out.print("Enter your age: ");

        // 3. Pause the program and wait for the user to type an integer and hit Enter
        int age = keyboard.nextInt();

        System.out.println("Next year you will be: " + (age + 1));

        // 4. Always close the scanner at the end of your program to clean up memory
        keyboard.close();
    }
}
```